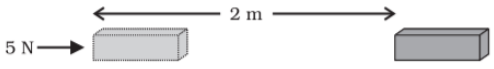


SOUND-02

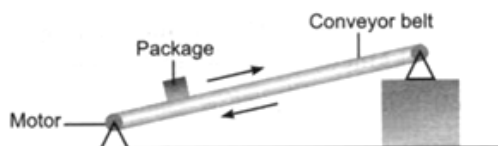
Class 09 - Science

1. What is negative work. [1]
 2. A boy of mass 50 kg runs up a staircase of 45 steps in 9 s. If the height of each step is 15 cm, find his power. [1]
Take $g = 10 \text{ ms}^{-2}$.
 3. What type of energy is stored in the spring of a watch? [1]
 4. Define 1 J or work. [1]
 5. Name the two common forms of mechanical energy. [1]
 6. Name at least three commonly used units of energy. [1]
 7. A person holds a bundle of hay over his head for 30 minutes and gets tired. Has he done some work or not? [1]
 8. When a book is lifted from a table, against which force work is done? [1]
 9. A force of 5 N is acting on an object. The object is displaced through 2 m in the direction of the force (Fig.). If the force acts on the object all through the displacement, then work done is $5 \text{ N} \times 2 \text{ m} = 10 \text{ N m}$ or 10 J. [1]
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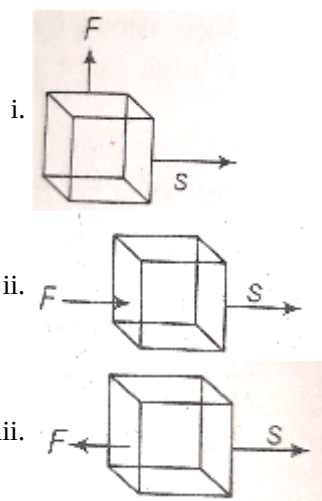
The diagram shows a horizontal force of 5 N acting to the right on a rectangular block. A horizontal arrow above the block indicates a displacement of 2 m to the right, in the same direction as the force.
10. Soni says that the acceleration in an object could be zero even when several forces are acting on it. Do you agree with her? Why? [1]
 11. Is it possible that some force is acting on a body but still the work done is zero? [1]
 12. What are the various energy transformations that occur when you are riding a bicycle? [1]
 13. Can there be displacement of an object in the absence of any force acting on it? Think. Discuss this question with your friends and teacher. [1]
 14. What is power? [1]
 15. A freely falling object eventually stops on reaching the ground. What happens to its kinetic energy? [1]
 16. What is the amount of work done by a force when a body moves in a circular path? [1]
 17. A body performs no work. Does it imply that the body possesses no energy? [1]
 18. An object of mass 15 kg is moving with a uniform velocity of 4 ms^{-1} . What is the kinetic energy possessed by the object? [1]
 19. Name at least six forms of energy. [1]
 20. Does the transfer of energy take place when you push a huge rock with all you might and fail to move it? Where is the energy you spend going? [1]
 21. What is energy? [1]
 22. What will cause greater change in kinetic energy of a body? Changing its mass or changing its velocity? [1]
 23. What will have happen to the kinetic energy of a body if its velocity is halved? [1]
 24. What is the kinetic energy of an object? [1]
 25. By how much will the speed of a body, of fixed mass, increase if its kinetic energy becomes four times its initial [1]

kinetic energy?

26. When do we say that work is done? [1]
27. Define 1 J of work. [1]
28. Find the energy possessed by an object of mass 10 kg when it is at a height of 6 m above the ground. Given, $g = 9.8 \text{ m s}^{-2}$. [1]
29. What is the work done when the force acting in the body and the displacement produced in the body are at right angles to each other? [1]
30. Define work. How is work measured? When is work done by a force negative? [1]
31. An electric bulb of 60 W is used for 6 h per day. Calculate the units of energy consumed in one day by the bulb. [1]
32. In a tug-of-war one team gives way to the other. What work is being done and by whom? [1]
33. An object of mass 12 kg is at a certain height above the ground. If the potential energy of the object is 480 J, find the height at which the object is with respect to the ground. Given, $g = 10 \text{ ms}^{-2}$. [1]
34. Define 1 watt of power. [1]
35. A porter lifts luggage of 15 kg from the ground and puts it on his head 1.5 m above the ground. Calculate the work done by him on the luggage. [1]
36. A pair of bullocks exerts a force of 140 N on a plough. The field being ploughed is 15 m long. How much work is done in ploughing the length of the field? [1]
37. A lamp consumes 1000 J of electrical energy in 10 s. What is its power? [1]
38. i. The potential energy of a freely falling object decreases progressively. What happens to its [3]
a. Kinetic energy,
b. total mechanical energy?
State the law on which your answer is based.
- ii. A household consumes 1 kWh of energy per day. How much energy is this in joules?
39. Figure shows a conveyor belt transporting a package to a raised platform. The belt is driven by a motor. [3]



- i. State three types of energy, other than gravitational potential energy, into which the electrical energy supplied to the motor is converted.
- ii. The mass of the package is 36 kg. Calculate the increase in the gravitational potential energy (p.e.) of the package when it is raised through a vertical height of 2.4 m.
- iii. The package is raised through the vertical height of 2.4 m in 4.4 s. Calculate the power needed to raise the package.
- iv. Assume that the power available to raise package is constant. A package of mass greater than 36 kg is raised through the same height. Suggest explain the effect of this increase in mass on the operation of the belt.
40. In each of the following, a force F is acting on an object of mass m . The direction of displacement is from West to east shown by the longer arrow. Observe the figure carefully and state whether the work done by the force is negative, positive or zero. [3]



41. A test tube loaded with lead shots weighs 50 gf and floats upto the mark 'X' in water. The test tube is then made to float alcohol. It is found that 10 gf of lead shots have to be removed, so as to float it to level 'X'. Find RD of alcohol. [3]

42. Ram's family was worried about heavy electricity bills to be paid. Their neighbour Mohan suggested some easy and effective steps to reduce the same. Next month's bill came as a relief to Ram, as the consumption of electricity had reduced by 50 units and so had the bill. [3]

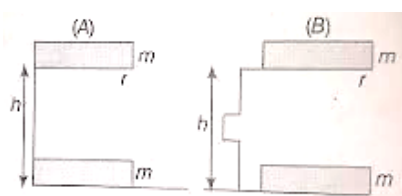
- In what other aspects of life can this situation help?
- What is the unit of energy?
- Write any three steps that you think Mohan might have suggested to Ram.

43. Shyam and his friends were playing with a catapult (gulel) in his garden. Several mangoes were dislodged and fell with the help of catapult. One of his friend was aiming the catapult on a bird. Shyam prevented him from doing so. [3]

- Name the energy possessed by the stretched string of the catapult.
- What will happen if the stone is thrown without stretching the string of a catapult?
- Why did Shyam prevent his friend from aiming at the bird? Which quality is highlighted in Shyam's behaviour?

44. Prachi and Kanchi were observing a building having two different staircase. One slanting and other vertically spiral. Prachi was of the opinion that a person using slanting staircase will be doing more work against gravity but Kanchi thought otherwise. They started arguing. Sanchi, their friend, explained and gave entirely different view and pacified them. [3]

- What according to you was the explanation given by Sanchi?
- What appreciable values do you see in the Sanchi?



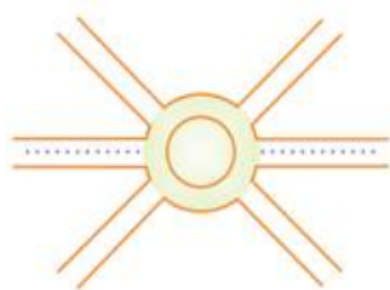
45. [3]

- What is meant by potential energy of a body?
- A body of mass m is raised to a vertical height h through two different paths A and B.

What will be the potential energy of the body in the two cases? Give reason for your answer.

46. Sony says that the acceleration in an object could be zero even when several forces are acting on it. Do you agree with her? Why? [3]

47. On a railway station, a passenger hires a coolie to pick his luggage and carry it to the train. When the coolie finished his work, he asked for the money. But the passenger refused to pay him saying that he has done no work. Answer the following questions based on the above information: [3]
- Was the passenger right in saying so?
 - What values are not promoted by the passenger?
 - Give a situation seen in everyday life when no work is said to be done by you even though you get tired.
48. An automobile engine propels a 1,000 kg car A along a levelled road at a speed of 36 km h^{-1} . Find the power if the opposing frictional force is 100 N. Now, suppose after travelling a distance of 200 m, this car collides with another stationary car B of same mass and comes to rest. Let its engine also stop at the same time. Now, car B starts moving on the same level road without getting its engine started. Find the speed of the car B just after the collision. [3]
49. A boy is moving on a straight road against a frictional force of 5 N. After travelling a distance of 1.5 km he forgot the correct path at a round about (Fig.) of radius 100 m. However, he moves on the circular path for one and half cycle and then he moves forward upto 2.0 km. Calculate the work done by him. [3]



50. Calculate the force of gravitation between the earth and the sun, given that the mass of the earth = $6 \times 10^{24} \text{ kg}$ and of the sun = $2 \times 10^{30} \text{ kg}$. The average distance between the two is $1.5 \times 10^{11} \text{ m}$. [3]
51. Due to heavy rains and speedy winds a tree broke and fell on the road. Two persons from local shops came and begin to remove that broken part of the tree, but in vain as the log was too heavy. A passerby came to their help with a few solid heavy sticks. Consequently, the heavy log was put aside and the commutation on the road began as usual. [3]
- Comment upon the value(s) displayed by the passerby?
 - Name the energy stored when the heavy log of the tree was lifted to some height?
52. Anil lives in a village and his school is 8 km away from his home. His father suggests buying a motor cycle to go to school. Anil opposes the idea and opts for a bicycle instead. [3]
- Write the energy transformation taking place while Anil rides his bicycle.
 - Justify (the stand taken by Anil in your own words).
 - How can he convince his friends to do the same?
53. Give reason : An iron nail sinks in water, but a ship made of iron floats. [3]
54. Amit buys few grams of gold at the poles as per the instruction of one of his friends. He hands over the same when he meets him at the equator. Will the friend agree with the weight of gold bought? If not, why? [3]
55. Avinash can run at a speed of 8 ms^{-1} against the frictional force of 10 N, and Kapil can move at a speed of 3 ms^{-1} against the frictional force of 25 N. Who is more powerful and why? [3]
56.
 - What do you mean by work? Give an example of negative work done.
 - Can a body have energy, without having momentum? If yes, why? [5]
57. Illustrate the law of conservation of energy by discussing the energy changes which occur when we draw a [5]

- pendulum bob to one side and allow it to oscillate. Why does the bob eventually come to rest? What happens to its energy eventually? Is it a violation of the law of conservation of energy?
58. Derive an expression for the kinetic energy of the body? Calculate the kinetic energy for a body of mass 5 Kg moving a velocity 2.5 m/s^2 [5]
59. A man standing on the top of a tower 60 m high throws a ball in the vertically upward direction with a velocity of 20 ms^{-1} . How long will it take the ball to pass by the man while moving in the downward direction? How long will it take the ball to hit the ground? (Take $g = 10 \text{ ms}^{-2}$) [5]
60. A car is moving with uniform velocities: 18 km h^{-1} , 36 km h^{-1} , 54 km h^{-1} and 72 km h^{-1} at some intervals. Find the KE of the boy of 40 kg sitting in the car at these velocities. Draw a graph between the KE and the velocities. Also, find the nature of the curve. [5]
61. i. Define work. State two factors on which the magnitude of work depends. [5]
ii. A car and a truck have the same speed of 30 m/s. If their masses are in the ratio 2 : 3; find the ratio of kinetic energy.
62. i. A body has a mass m and velocity v . If the mass is increased four times and velocity is decreased two times, calculate the ratio of the kinetic energies in the above cases. [5]
ii. What kind of energy transformation takes place when sparkle is lighted?
63. a. Give one example of each of the following: [5]
i. Small mass but high kinetic energy
ii. Large mass but low kinetic energy
b. Prove mathematically that the total mechanical energy of a freely falling body in air is conserved.
64. i. State the law of conservation of energy. [5]
ii. Illustrate the law of conservation of energy by discussing the energy changes which occur when we draw a pendulum bob to one side and allow it to oscillate.
65. Define watt. Express kilowatt in terms of joule per second. A 150-kg car engine develops 500 W for each kg. What force does it exert in moving the car at a speed of 20 ms^{-1} ? [5]
66. Briefly describing the gravitational potential energy, deduce an expression for the gravitational potential energy of a body of mass m placed at a height h , above the ground. [5]
67. Show that total energy is conserved if the ball of mass ' m ' is the thrown downwards from a height ' h '. [5]
68. i. What happens when you shake a wet piece of cloth? Explain your observation. [5]
ii. An electric bulb of 200 W is used for 6 hrs a day. Calculate the energy consumed by it in a day in Joules and kilowatt-hour unit.
69. Look at the activities listed below. Reason out whether or not work is done in the light of your understanding of the term 'work'. [5]
a. Suma is swimming in a pond.
b. A donkey is carrying a load on its back.
c. A wind-mill is lifting water from a well.
d. A green plant is carrying out photosynthesis.
e. An engine is pulling a train.
f. Food grains are getting dried in the sun.

g. A sailboat is moving due to wind energy.

70. i. Define kinetic energy of an object. Can kinetic energy of an object be negative? Give reason. [5]
ii. A car weighing 1200 kg is uniformly accelerated from rest and covers a distance of 40m in 5 s. Calculate the work the car engine had to do during this time.
71. The area of cross-section of the stem of a hydrometer is 0.32 cm^2 and its length is 20 cm. Its mass is 22.8 g and volume of the bulb is 10 cm^3 . [5]
Find.
i) The density of the liquid in which it floats upto its midpoint.
ii) The density of the liquid in which it is completely immersed.
72. Calculate the electricity bill amount for a month of April, if 4 bulbs of 40 W for 5 h, 4 tubelights of 60 W for 5 h, a TV of 100 W for 6 h, a washing machine of 400 W for 3 h are used per day. The cost per unit is Rs. 1.80. [5]
73. Show that when a body is dropped from a certain height, the sum of its kinetic energy at any instant during its fall is constant. [5]